

Llista 1. Estadística Descriptiva. Solucions

Exercici 1

Fills	Freq. abs	Freq. abs. acum.	Freq. rel.	Freq. rel. acum.
0	52	52	0.26	0.26
1	48	100	0.24	0.5
2	45	145	0.225	0.725
3	25	170	0.125	0.85
4	28	198	0.14	0.99
5++	2	200	0.1	1

Exercici 2

Notes	Freq. abs	Freq. abs. acum.	Freq. rel.	Freq. rel. acum.
0	4	4	0.1	0.1
1	4	8	0.1	0.2
2	5	13	0.125	0.325
3	4	17	0.1	0.425
4	6	23	0.15	0.575
5	6	29	0.15	0.725
6	3	32	0.075	0.8
7	3	35	0.075	0.875
8	3	38	0.075	0.95
9	1	39	0.025	0.975
10	1	40	0.025	1

```
> x <- c(7, 3, 2, 4, 5, 1, 8, 6, 1, 5, 3, 2, 4, 9, 8, 1, 0, 2, 4, 1, 2, 5, 6, 5, 4, 7, 1, 3,
```

```
## b)
```

```
> mean(x)
```

```
[1] 4.075
```

```
> sd(x)*(length(x)-1)/length(x)      # Desviació típica
```

```
[1] 2.601802
```

```
> median(x)
```

```
[1] 4
```

Moda = 4 i 5.

Exercici 3

Interval	Marca	Freq. abs	Freq. abs. acum.	Freq. rel.	Freq. rel. acum.
[0,5)	2.5	10	10	0.32	0.32
[5,7)	6	8	18	0.26	0.58
[7,10)	8.5	9	27	0.29	0.87
[10,12]	11	4	31	0.13	1

a) Q1 $\rightarrow$ [0,5) Q2 $\rightarrow$ [5,7) Q3 $\rightarrow$ [7,10)

b) $Me = L_i + (L_{i+1} - L_i) \frac{\frac{n}{2} - N_{i-1}}{n_i} = 5 + (7 - 5) \frac{\frac{31}{2} - 10}{8} = 6.375$

c) 20%-percentil $\rightarrow$ [0,5) 85%-percentil $\rightarrow$ [7,10)

Exercici 4

```
> stol <- c(271,428,381,366,411,193,178,178,427,180)
```

```
## a)
```

```
> sort(stol)
[1] 178 178 180 193 271 366 381 411 427 428
> median(stol)
[1] 318.5
> table(stol)
stol
178 180 193 271 366 381 411 427 428
  2   1   1   1   1   1   1   1   1
```

```
## b)
```

```
> var(stol); sd(stol)
[1] 12465.79
[1] 111.6503
```

c) $y_i = kx_i$

Mitjana:

$$\bar{y} = \sum_{i=1}^n y_i = \sum_{i=1}^n kx_i = k \sum_{i=1}^n x_i = k\bar{x}$$

Variància:

$$s_y^2 = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (kx_i - k\bar{x})^2 = k^2 \sum_{i=1}^n (x_i - \bar{x})^2 = k^2 s_x^2$$

```
> k <- 3.28084
> mean(stol)*k
[1] 988.5171
> var(stol)*k^2
[1] 134180.6
```

Exercici 5

```
## a)
> # Rang
> max(temperatura)-min(temperatura)
[1] 26.5
> # Rang interquantil.lic
> summary(temperatura)[5]-summary(temperatura)[2]
3rd Qu.
6.875
```

b)

c) Número de classes: 7

Longitud dels intervals: 4 °C

d)

Temper. (Marca)	Freq. abs	Freq. abs. acum.	Freq. rel.	Freq. rel. acum.
[65,69)	1	1	0.033	0.033
[69,73)	5	6	0.167	0.2
[73,77)	4	10	0.133	0.333
[77,81)	7	17	0.233	0.566
[81,85)	8	25	0.267	0.833
[85,89)	3	28	0.1	0.933
[89,93]	2	30	0.067	1

```
## e)
> (7/30)*3/4
[1] 0.175 # 17.5%
```

Exercici 6

	N	SN	MSN	
Impur	19	29	24	72
No impur	560	269	497	1326
	579	298	521	1398

a) $72/1398 = 0.0515 = 5.15\%$

b) $579/1398 = 0.4142 = 41.42\%$

c) $497/1398 = 0.3555 = 35.55\%$

d) $(29+24)/1398 = 0.0379 = 3.79\%$

Exercici 7

```
> eritrocits <- c(4.2, 5.7, 6.1, 3.8, 4.5, 5.2, 4.6, 4.3)
> ferro <- c(33, 48, 53, 44, 41, 39, 42, 36)

## a)
> mean(eritrocits); mean(ferro)
[1] 4.8
[1] 42

## b)
> var(eritrocits); var(ferro)
[1] 0.6285714
[1] 41.14286
> sd(eritrocits); sd(ferro)
[1] 0.792825
[1] 6.41427

## c)
> # Coeficient de variació
> coef_var(eritrocits); coef_var(ferro)
[1] 0.1651719
[1] 0.1527207

> ## d)
> cov(eritrocits,ferro)
[1] 3.6
> cor(eritrocits,ferro)
[1] 0.7079099
```

Exercici 8

```
a) 16 anys - Enunciat: dim(previsions)
b) Enunciat: mean(...), var(...), cov(...)

## c)
cor(Demanda, Vendes)
[1] 0.8194537
# Correlació lineal moderada positiva

## d)
> reg

Call:
```

```

lm(formula = Vendes ~ Demanda)

Coefficients:
(Intercept)      Demanda
      1.43558       0.02354
# També es pot trobar a l'enunciat: summary(reg) - Coefficients

## e)
> new.x <- data.frame(Demanda = c(150, 210, 380))
> predict(reg, newdata = new.x)
      1      2      3
4.966077 6.378275 10.379502

## f)
# Valors ajustats
> v_ajustats <- predict(reg)
> v_ajustats[1:3]
      1      2      3
6.142909 6.613641 10.850235
# Residus
> residus <- Vendes-v_ajustats
> residus[1:3]
      1      2      3
2.8570913 -0.6136413  1.1497654

```

Exercici 9

a) Enunciat: mean(...), var(...), cov(...)

Unitats de mesura: Mitjanes g i kcal, variàncies g² i kcal², covariància g · kcal.

```

## b)
cor(carbohidrats, calories)
[1] 0.7247129
# Correlació lineal moderada positiva

```

c)

```
> reg
```

Call:

```
lm(formula = calories ~ carbohidrats)
```

Coefficients:

```

(Intercept)  carbohidrats
      93.00       10.65

```

Coefficient de determinació: 0.4659 → 46.6% - Enunciat: Adjusted R - squared.

```
## d)
> new.x <- data.frame(carbohidrats = 5)
> predict(reg, newdata = new.x)
      1
146.2287
```

Exercici 10

```
> pares <- c(165, 160, 170, 162, 173, 157, 178, 168, 173, 170)
> fills <- c(173, 168, 173, 167, 175, 168, 173, 170, 174, 170)
```

```
> ## a)
> mean(pares); mean(fills)
[1] 167.6
[1] 171.1
> var(pares); var(fills)
[1] 42.93333
[1] 8.1
> sd(pares); sd(fills)
[1] 6.552353
[1] 2.84605
```

```
## b)
# Covariancia
> cov(pares, fills)
[1] 14.71111
# Coeficient de correlacio
> cor(pares, fills)
[1] 0.7888704
# Coeficient de determinacio
> cor(pares, fills)^2
[1] 0.6223165
```

```
## c)
> reg <- lm(fills~pares)
> reg
```

Call:

```
lm(formula = fills ~ pares)
```

Coefficients:

```
(Intercept)      pares
    113.6718      0.3427
```

```
## d)
> new.p <- data.frame(pares = 166)
```

```
> predict(reg, newdata = new.p)
      1
170.5518
```

Exercici 11

```
> x <- c(1, 2, 6, 3, 5, 10)
> y <- c(3.8, 8.2, 19.1, 9.5, 15, 31.1)
```

a)

```
> reg <- lm(y~x)
> reg
```

Call:

```
lm(formula = y ~ x)
```

Coefficients:

```
(Intercept)          x
      1.064         2.975
```

b)

Covariancia

```
> cov(x, y)
```

```
[1] 31.83
```

Coeficient de correlacio

```
> cor(x, y)
```

```
[1] 0.9971597
```

Coeficient de determinacio

```
> cor(x, y)^2
```

```
[1] 0.9943274
```

c)

Valors ajustats

```
> y_ajustats <- predict(reg)
```

```
> y_ajustats
```

```
      1      2      3      4      5      6
4.038318 7.013084 18.912150 9.987850 15.937383 30.811215
```

Residus

```
> residus <- y-y_ajustats
```

```
> residus
```

```
      1      2      3      4      5      6
-0.2383178 1.1869159 0.1878505 -0.4878505 -0.9373832 0.2887850
```